Chapter 16. Logistic Regression

Logistic model, odds ratios, maximum likelihood and IRLS, 2×2 tables

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16.1 A model for a binary response

When y takes only the values 0 and 1, $E(y \mid x) = P(y = 1 \mid x) =: \pi(x)$ is a probability. The linear model $y = \beta_0 + \beta_1 x + \epsilon$ is unsuitable: fitted values can leave $[0, 1]$, the errors take only two values for each x (so they are not normal), and $\mathrm{Var}(y \mid x) = \pi(x)\{1 - \pi(x)\}$ varies with x. Empirically, $\pi(x)$ often follows an S-shaped curve. We therefore model

$$\pi(x) = G(\beta_0 + \beta_1 x)$$

with a cdf G, the inverse **link function**. Common choices: the logistic $G(t) = e^t/(1 + e^t)$, the normal cdf Φ (probit), and the Gumbel $G(t) = \exp(-e^{-t})$. The logistic choice gives the **logistic regression model**

$$\pi(x) = \frac{\exp(\beta_0 + \beta_1 x)}{1 + \exp(\beta_0 + \beta_1 x)}, \qquad \text{equivalently} \qquad \operatorname{logit} \pi(x) = \log \frac{\pi(x)}{1 - \pi(x)} = \beta_0 + \beta_1 x.$$

The curve increases in x if $\beta_1 > 0$ and decreases if $\beta_1 < 0$.

Odds and odds ratios. The **odds** of $y = 1$ are $\pi/(1 - \pi) = P(y = 1)/P(y = 0) \in [0, \infty)$, with $\pi = \text{odds}/(1 + \text{odds})$. Under the logistic model

$$\frac{\pi(x)}{1 - \pi(x)} = e^{\beta_0}(e^{\beta_1})^x, \qquad \frac{\pi(x+1)/\{1 - \pi(x+1)\}}{\pi(x)/\{1 - \pi(x)\}} = e^{\beta_1}.$$

So e^{β_1} is the **odds ratio** for a unit increase in x, the same at every x. For a binary x (exposed $= 1$), e^{β_1} is the odds ratio between the two groups; for instance $\hat\beta_1 = 2.0$ means the odds of $y = 1$ are $e^2 \approx 7.4$ times higher in the exposed group.

16.2 Estimation

Motivation. For a binary response the linear model fails (fitted probabilities outside $[0, 1]$, nonconstant variance). Logistic regression models the log-odds linearly, and its coefficients are estimated by maximum likelihood with iterative reweighted least squares.

With $\mathbf{x}_i = (1, x_{i1}, \ldots, x_{ip})'$ and $\pi_i = \pi(\mathbf{x}_i) = e^{\mathbf{x}_i'\beta}/(1 + e^{\mathbf{x}_i'\beta})$, independent observations $y_i \sim$ Bernoulli(π_i) give the likelihood $L(\beta) = \prod \pi_i^{y_i}(1 - \pi_i)^{1-y_i}$ and

$$l(\beta) = \sum_{i=1}^{n} \{y_i \log \pi_i + (1 - y_i) \log(1 - \pi_i)\} = \sum_{i=1}^{n} \{y_i \mathbf{x}_i'\beta - \log(1 + e^{\mathbf{x}_i'\beta})\}.$$

Theorem 16.1

With $\pi(\beta) = (\pi_1, \ldots, \pi_n)'$ and $W(\beta) = \mathrm{diag}\{\pi_i(1 - \pi_i)\}$,

$$\frac{\partial l}{\partial \beta} = X'\{\mathbf{y} - \pi(\beta)\}, \qquad \frac{\partial^2 l}{\partial \beta \partial \beta'} = -X'W(\beta)X.$$

The Hessian is negative definite (if X has full column rank), so l is strictly concave and the MLE, when it exists, is the unique root of $X'\{\mathbf{y} - \pi(\hat\beta)\} = \mathbf{0}$.

Proof. $\frac{d}{dt}\log(1 + e^t) = \frac{e^t}{1+e^t}$ and $\frac{d^2}{dt^2}\log(1 + e^t) = \frac{e^t}{(1+e^t)^2} = G(t)\{1 - G(t)\}$; apply the chain rule with $t_i = \mathbf{x}_i'\beta$. Since $0 < \pi_i < 1$, $\mathbf{a}'X'WX\mathbf{a} = \sum_i \pi_i(1 - \pi_i)(\mathbf{x}_i'\mathbf{a})^2 > 0$ for $\mathbf{a} \neq \mathbf{0}$. ∎

The likelihood equations are nonlinear and are solved by Newton–Raphson:

$$\beta^{(t+1)} = \beta^{(t)} + \{X'W^{(t)}X\}^{-1}X'\{\mathbf{y} - \pi^{(t)}\} = \{X'W^{(t)}X\}^{-1}X'W^{(t)}\mathbf{z}^{(t)},$$

with the **working response**

$$\mathbf{z}^{(t)} = X\beta^{(t)} + \{W^{(t)}\}^{-1}\{\mathbf{y} - \pi^{(t)}\}.$$

Each step is a weighted least squares regression of $\mathbf{z}^{(t)}$ on X with weights $\pi_i^{(t)}(1-\pi_i^{(t)})$:

$$\beta^{(t+1)} = \arg\min_{\beta}\{\mathbf{z}^{(t)} - X\beta\}'W^{(t)}\{\mathbf{z}^{(t)} - X\beta\},$$

hence the name **iteratively reweighted least squares** (IRLS). A common start is $\beta^{(0)} = \mathbf{0}$. Since the Hessian does not depend on $\mathbf{y}$, Newton–Raphson coincides with Fisher scoring, and $\{X'W(\hat\beta)X\}^{-1}$ estimates the asymptotic covariance of $\hat\beta$. Wald tests and intervals use $\hat\beta_j/\mathrm{se}(\hat\beta_j) \approx N(0,1)$; likelihood ratio tests compare $2\{l(\hat\beta_F) - l(\hat\beta_R)\}$ with a chi-square distribution.

If the two classes can be perfectly separated by a hyperplane in x, l increases without bound along a direction and the MLE does not exist.

16.3 Two-by-two tables

Motivation. With a single binary predictor, logistic regression reduces to a 2×2 table, and its slope is the log odds ratio. This connects the model with classical contingency-table analysis.

Binary exposure (x) and outcome (y) data arise from three designs:

- **cohort** (prospective) studies follow exposed and unexposed groups and record outcomes;
- **case–control** (retrospective) studies sample cases ($y = 1$) and controls ($y = 0$) and record past exposure;
- **cross-sectional** studies sample individuals and record both at one time.

With counts n_{xy}:

	$y=0$	$y=1$	total
$x=0$	n_{00}	n_{01}	n_{0+}
$x=1$	n_{10}	n_{11}	n_{1+}

The **odds ratio** is $OR = \dfrac{\pi_1/(1-\pi_1)}{\pi_0/(1-\pi_0)}$ with $\pi_x = P(y=1 \mid x)$, estimated by $\widehat{OR} = \dfrac{n_{11}n_{00}}{n_{10}n_{01}}$.

Theorem 16.2

In the logistic model $\mathrm{logit}\, P(y=1 \mid x) = \beta_0 + \beta_1 x$ with binary x, the MLEs satisfy

$$\hat\pi_1 = \frac{e^{\hat\beta_0+\hat\beta_1}}{1+e^{\hat\beta_0+\hat\beta_1}} = \frac{n_{11}}{n_{1+}}, \qquad \hat\pi_0 = \frac{e^{\hat\beta_0}}{1+e^{\hat\beta_0}} = \frac{n_{01}}{n_{0+}}, \qquad e^{\hat\beta_1} = \frac{n_{11}n_{00}}{n_{10}n_{01}}.$$

Proof. The log-likelihood is

$$l = n_{11}(\beta_0+\beta_1) + n_{01}\beta_0 - n_{1+}\log(1+e^{\beta_0+\beta_1}) - n_{0+}\log(1+e^{\beta_0}).$$

The model is saturated (two parameters for two probabilities), so by invariance the MLEs of π_1, π_0 are the sample proportions n_{11}/n_{1+} and n_{01}/n_{0+}. Then $e^{\hat\beta_1} = \frac{\hat\pi_1/(1-\hat\pi_1)}{\hat\pi_0/(1-\hat\pi_0)} = \frac{n_{11}/n_{10}}{n_{01}/n_{00}}$. ∎

Why the odds ratio. In a case–control study the row proportions do not estimate π_x, because the numbers of cases and controls are fixed by design. But the odds ratio is symmetric in x and y:

$$\frac{P(y=1 \mid x=1)P(y=0 \mid x=0)}{P(y=0 \mid x=1)P(y=1 \mid x=0)} = \frac{P(x=1 \mid y=1)P(x=0 \mid y=0)}{P(x=0 \mid y=1)P(x=1 \mid y=0)},$$

since both equal $\frac{p_{11}p_{00}}{p_{10}p_{01}}$ in terms of the joint cell probabilities. So the odds ratio, and hence the slope β_1 of a logistic regression, can be estimated from case–control data; only the intercept is distorted by the sampling.

Example 16.1

In a trial, 140 subjects receive a supplement and 140 a placebo; 17 and 31 catch a cold, respectively. With $x = 1$ for the supplement, $\widehat{OR} = \frac{17 \cdot 109}{123 \cdot 31} = 0.49$: the odds of a cold are about halved, and $\hat{\beta}_1 = \log 0.49 = -0.72$. The standard error is $\sqrt{\frac{1}{17} + \frac{1}{123} + \frac{1}{31} + \frac{1}{109}} = 0.33$, so the Wald statistic is -2.19, significant at the 5% level.